



# 2232 - Precision Tests of the Physics of Mixing in Cool Planetary and Brown Dwarf Atmospheres

Cycle: 1, Proposal Category: AR

## INVESTIGATORS

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## OBSERVATIONS

## ABSTRACT

Vertical mixing in giant planet and brown dwarf atmospheres controls crucial aspects of cloud thickness and particle sizes, the transport of molecules into the stratosphere, and deviations from equilibrium chemistry. Unfortunately, the fundamental atmospheric parameter used in modeling these processes, the vertical eddy diffusion coefficient ( $K_{zz}$ ) is unknown to within many orders of magnitude. However, it is now apparent that precision atmospheric abundances of non-equilibrium chemical species, which will be delivered by JWST spectra, can be used to constrain  $K_{zz}$  to a level not previously appreciated. At the same time, a variety of new 1D and 3D theories have been put forth to predict  $K_{zz}$  in radiative regions, to complement mixing length theory in convective regions. We aim to put the breadth of these theories to the test by coupling them to a 1D radiative-equilibrium atmosphere code to predict non-equilibrium atmospheric abundances and spectra that can be directly compared to forthcoming JWST spectroscopy of transiting planets, imaged planets, and brown dwarfs. The outcome of the synergy of JWST observations with these models will be orders of magnitude better constraints on  $K_{zz}$ , by far the most poorly known parameter in giant planet and brown dwarf atmospheres, enabling significantly better interpretation of JWST observations.

## **OBSERVING DESCRIPTION**

This is a Theory Proposal.